CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 17: Circles

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Cyclic quadrilaterals

1. A, E, C and B lie in order on a circle. Chords AC and BC are equal, BC is produced to D, and $\angle ACD = 120°$. Find $\angle AEC$. (a) $30°$ (b) $60°$ (c) $90°$ (d) $120°$ ICSE 2025, 1 mark

SOLUTION

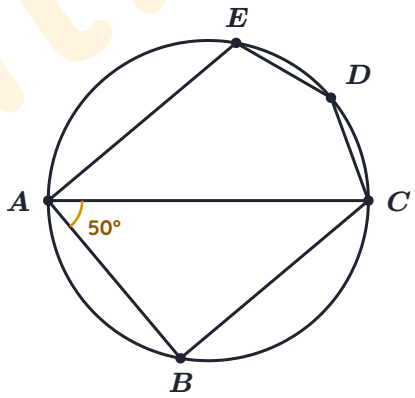
$\angle ACB = 180° - 120° = 60°$ (linear pair).

$AC = BC$, so $\angle ABC = \angle BAC = \dfrac{180° - 60°}{2} = 60°$.

$ABCE$ is a cyclic quadrilateral, so $\angle AEC = 180° - \angle ABC = 120°$.

Answer: (d)

2. $ABCDE$ is a pentagon inscribed in a circle. AC is a diameter, $BC \parallel AE$ and $\angle BAC = 50°$. Find (a) $\angle ACB$ (b) $\angle EDC$ (c) $\angle BEC$. Then prove that BE is also a diameter. ICSE 2019, 4 marks



SOLUTION

(a) $\angle ABC = 90°$ (angle in a semicircle), so $\angle ACB = 90° - 50° = 40°$.

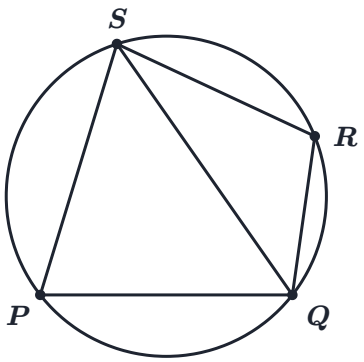
(b) $BC \parallel AE$, so $\angle EAC = \angle ACB = 40°$ (alternate angles). $ACDE$ is a cyclic quadrilateral, so $\angle EDC = 180° - 40° = 140°$.

(c) $\angle BEC = \angle BAC = 50°$ (angles in the same segment).

$\angle EAB = \angle EAC + \angle CAB = 40° + 50° = 90°$. A chord that subtends a right angle at the circumference is a diameter.

Answer: (a) $40°$ (b) $140°$ (c) $50°$; since $\angle EAB = 90°$, BE is a diameter

3. $PQRS$ is a cyclic quadrilateral with $\angle QPS = 73°$, $\angle PQS = 55°$ and $\angle PSR = 82°$. Find (a) $\angle QRS$ (b) $\angle RQS$ (c) $\angle PRQ$. ICSE 2018, 4 marks



SOLUTION

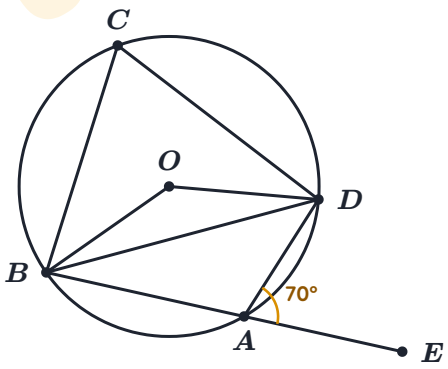
(a) Opposite angles of a cyclic quadrilateral add up to $180°$: $\angle QRS = 180° - 73° = 107°$.

(b) In $\triangle PQS$: $\angle PSQ = 180° - 73° - 55° = 52°$, so $\angle QSR = 82° - 52° = 30°$. In $\triangle QRS$: $\angle RQS = 180° - 107° - 30° = 43°$.

(c) $\angle PRQ = \angle PSQ = 52°$ (angles in the same segment).

Answer: (a) $107°$ **(b)** $43°$ **(c)** $52°$

4. $ABCD$ is a cyclic quadrilateral in a circle with centre O. BA is produced to E and $\angle DAE = 70°$. Find (a) $\angle BCD$ (b) $\angle BOD$ (c) $\angle OBD$. ICSE 2017, 4 marks



SOLUTION

(a) The exterior angle of a cyclic quadrilateral equals the interior opposite angle: $\angle BCD = \angle DAE = 70°$.

(b) The angle at the centre is twice the angle at the circumference: $\angle BOD = 2 \times 70° = 140°$.

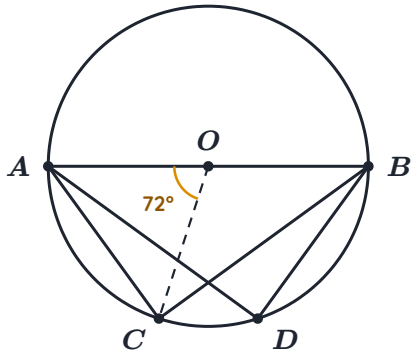
(c) $OB = OD$ (radii), so $\angle OBD = \dfrac{180° - 140°}{2} = 20°$.

Answer: (a) $70°$ **(b)** $140°$ **(c)** $20°$

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2 Angle at the centre and at the circumference

5. AB is a diameter of a circle with centre O. C and D are points on the circle on the same side of AB, with $AC = BD$ and $\angle AOC = 72°$. Find (a) $\angle ABC$ (b) $\angle BAD$ (c) $\angle ABD$. ICSE 2020, 3 marks



SOLUTION

(a) The angle at the circumference is half the angle at the centre on the same arc: $\angle ABC = \frac{72°}{2} = 36°$.

(b) Equal chords subtend equal angles at the centre, so $\angle BOD = 72°$ and $\angle BAD = 36°$.

(c) $\angle ADB = 90°$ (angle in a semicircle), so $\angle ABD = 90° - 36° = 54°$.

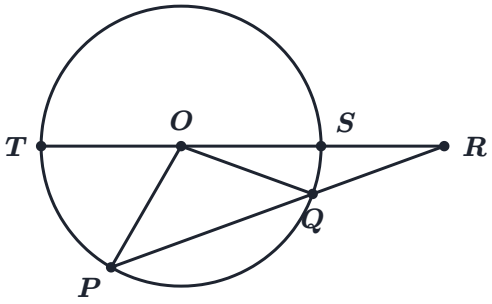
Answer: (a) $36°$ **(b)** $36°$ **(c)** $54°$

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3 Arcs, chords and angles

6. O is the centre of a circle. A line through O meets the circle at T and S and is produced to R. A line from R cuts the circle at Q and P, with Q nearer to R. $QR = OP$ and $\angle ORP = 20°$. Find $x = \angle TOP$.

ICSE 2018, 3 marks



SOLUTION

$OQ = OP$ (radii) and $QR = OP$, so $OQ = QR$. In $\triangle OQR$: $\angle QOR = \angle QRO = 20°$.

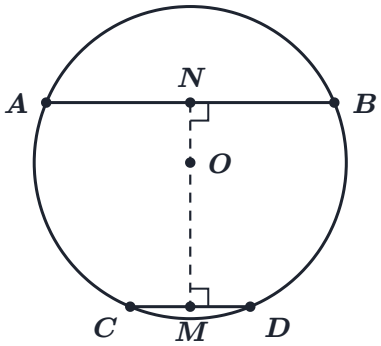
Exterior angle of $\triangle OQR$: $\angle OQP = 20° + 20° = 40°$.

$OP = OQ$, so $\angle OPQ = \angle OQP = 40°$.

x is an exterior angle of $\triangle OPR$: $x = \angle OPR + \angle ORP = 40° + 20°$.

Answer: $x = 60°$

7. AB and CD are parallel chords on opposite sides of the centre of a circle of radius 13 cm. $AB = 24$ cm and $CD = 10$ cm. Find the distance between the chords. ICSE 2017, 3 marks



SOLUTION

Draw $ON \perp AB$ and $OM \perp CD$. The perpendicular from the centre bisects a chord, so $AN = 12$ cm and $CM = 5$ cm.

$ON = \sqrt{13^2 - 12^2} = 5$ cm and $OM = \sqrt{13^2 - 5^2} = 12$ cm

The chords are on opposite sides of the centre, so the distance is $ON + OM$.

Answer: 17 cm

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